ADVANCED FUNCTIONAL MATERIALS
www.afm-journal.de

Space-Confined Yolk-Shell Construction of Fe_3O_4 Nanoparticles Inside N-Doped Hollow Mesoporous Carbon Spheres as Bifunctional Electrocatalysts for Long-Term Rechargeable Zinc–Air Batteries

*Bin Wang, Yuzhen Ye, Li Xu, Yu Quan, Wenxian Wei, Wenshuai Zhu, Huaming Li, and Jiexiang Xia**

Development of efficient, durable and inexpensive oxygen reduction reaction (ORR) and oxygen evolution reaction (OER) electrocatalysts with accelerated kinetics and high-performance remain a grand challenge in the context of reversible metal–air batteries. Herein, the Fe_3O_4 nanoparticles inside N-doped hollow mesoporous carbon spheres (N/HCSs) yolk-shell structure (Fe_x@N/HCSs) is constructed as an excellent bifunctional electrocatalyst for ORR and OER via an innovative approach. The N/HCSs effectively control and confine in situ growth of Fe_3O_4 nanoparticles using the melting-diffusion strategy via capillary force and significantly improve the conductivity and structural stability of the hybrid material. The constructed yolk-shell structured Fe_{20}@N/HCSs ecosystem with Fe–N_x active sites exhibits excellent ORR and OER activity and stability, which even surpass commercial grade Pt/C, RuO_2, IrO_2 and many reported catalysts. Moreover, the zinc–air battery assembled with Fe_{20}@N/HCSs as a cathode achieves high open circuit voltage (1.57 V), large power density (140.8 mW cm^{-2}), and excellent long-term cycling performance (over 300 h), revealing superior performance compared to commercial Pt/C + RuO_2. This work provides a new avenue for the design and optimization of other high-performance yolk-shell materials with nanoscale confinement structures.

1. Introduction

With the rapid development of economy, energy shortage and environmental pollution are becoming serious increasingly, thus the demand for clean renewable energy is increasing.[1,2] Metal–air batteries, especially zinc–air batteries, are considered as potential substitutes for lithium-ion batteries because of their environmentally friendly, low cost-effectiveness, safety, rechargeability, and excellent durability.[3,4] At present, platinum (Pt) and ruthenium (Ru) based materials are the most active catalysts for oxygen reduction reaction (ORR) and oxygen evolution reaction (OER), respectively.[5,6] However, due to the high cost of noble metal based catalysts, the development of zinc air batteries has been limited. The research and development of non-noble metal catalysts has become one of the research topics of zinc air batteries.[7,8]

In the field of ORR/OER, carbon based materials have attracted extensive attention due to the wide sources, excellent conductivity, large specific surface area, and excellent thermodynamic properties.[9,10] Meanwhile, the introduction of heteroatom nitrogen makes the carbon surface get more activated π electrons, which can change the local density state near the Fermi level of graphite carbon, reduce the reaction work function, and increase the density of active center, thus effectively promoting the electrocatalytic ORR/OER reaction.[11,12] In addition, the introduction of transition metals into nitrogen-doped carbon materials can further enhance the electrocatalytic ORR/OER activity of these catalyst.[13,14] For example, the introduction of iron can regulate the electronic structure of the catalyst so that the d-band shrinks, and the electron density increases closer to the Fermi level.[15,16] This accelerates the electrons transfer to the reaction intermediates, making the Fe–N active site more prone to electrocatalytic oxygen reaction. Therefore, the transition metal modified N-doped carbon has become the most potential material to replace the noble metal catalyst.[17,18]

Among many carbon-based materials, the yolk-shell structure carbon material with the distinctive core–void–shell configuration have become a hot research topic, which present many advantages at the same time, including the intriguing properties of cores, interstitial hollow spaces, and diverse

Dr. B. Wang, Y. Ye, Dr. L. Xu, Y. Quan, Prof. W. Zhu, Prof. H. Li, Prof. J. Xia
School of Chemistry and Chemical Engineering
Institute for Energy Research
Jiangsu University
Zhenjiang 212013, China
E-mail: xjx@ujs.edu.cn
Dr. W. Wei
Testing center
Yangzhou University
Yangzhou, Jiangsu 225009, China
The ORCID identification number(s) for the author(s) of this article can be found under https://doi.org/10.1002/adfm.202005834.

DOI: 10.1002/adfm.202005834

functionalities of shells.[19,20] The transition metal nanoparticles are confined within permeable robust N-doped carbon shell, which nanoscale void formed between yolk and shell to induce localized high instantaneous concentration for driving fast heterogeneous catalysis.[21,22] The N-doped mesoporous carbon shell can be used as an effective electron conductor, electrolyte transporter and yolk nucleus protector, which can effectively inhibit the agglomeration of transition metal nanoparticles and confine the catalytic ORR/OER in the void of yolk- shell structure.[23,24] So far, the design of yolk-shell structure carbon materials are mainly through layer-by-layer templating strategy.[25,26] With noble metal or metal oxide nanoparticles as the core, the silica template is first coated, and then the carbon source is wrapped. After the carbonization, the silica template is etched to obtain a yolk-shell structure carbon material. Alternatively, a metal oxide is used as the core, and after the carbon shell is constructed, a part of the core is etched to directly obtain a yolk-shell structure carbon material.[27,28] In general, because it is difficult to precisely control chemical reactions and material compatibility in confined nanospaces, the structure and composition of yolk-shell remains a major challenge.

Herein, with the special capillary effect of mesoporous carbon shell, the adsorbed iron ions are locked in the cavity of carbon sphere, and Fe_3O_4 nanoparticles grow on the inner wall of N-doped hollow carbon spheres (Fe_x@N/HCSs) after thermal condensation.[29,30] The solid mesoporous carbon spheres can effectively protect iron oxide nanoparticles from falling off and agglomeration, and also ensure the full circulation of electrolyte and oxygen molecules, making each core–void–shell configuration an independent and efficient ORR/OER active ecosystem. Unique material properties endow Fe_x@N/HCSs with high electrocatalytic activity toward ORR/OER in terms of low overpotential, fast kinetic reaction rate, high current density and excellent durability. Therefore, the zinc–air batteries constructed using Fe_{20}@N/HCSs as the air electrode reveal the high voltage (1.57 V), maximum power density (140.8 mW cm^{-2}), and super durable stability (300 h). Far more than many of the precious metal and transition metal based materials currently reported.

2. Results and Discussion

A schematic diagram for the preparation of the yolk-shelled structure for Fe_3O_4 nanoparticles confined within N-doped carbon spheres is shown in **Figure 1**a. First, the uniform monodisperse SiO_2 nanospheres with diameter of nearly 300 nm are synthesized as templates by the Stöber method (Figure S1, Supporting Information).[31] In Tris(hydroxymethyl)aminomethane buffer solution, with the increase of reaction time, the solution changed from white to brown black, which was attributed to the formation of polydopamine (Figure S2, Supporting Information). After calcining SiO_2@polydopamine sample, the surface of SiO_2 is coated with 8 nm thick pyrolytic dopamine (pyro-PDA) (Figure S3, Supporting Information). The SiO_2 template was etched by NH_4HF_2 to obtain nitrogen doped hollow carbon spheres precursor (N/HCSs-P) with ultrathin (≈8 nm) carbon layer and abundant mesoporous structure (Figure 1b and Figure S4, Supporting Information). The N/HCSs-P was calcined in N_2 atmosphere for 1 h to obtain N/HCSs (Figure S5a, Supporting Information). The N/HCSs-P serve as nanoreactors, and the Fe^{3+} permeate into the inner voids of porous N/HCSs-P via the melting–diffusion strategy with capillary force. In the secondary pyrolysis process, N/HCSs-P can control the confined growth of metal oxide nanoparticles inside the carbon sphere. Similar strategies are also used in other systems, such as yolk-shell MoS_2@C nanospheres,[29] ZIF-67@HCSs,[30] etc. According to the morphology characterization, it can be found that the particle size of metal oxide nanoparticles increases with the increase of iron source (Figure S5, Supporting Information and Figure 1c–f). As shown in the Figure 1c–f, when the ratio of N/HCSs-P to $FeCl_3 \cdot 6H_2O$ is 20 mg:20 mg, the diameter of nanoparticles growing on the inner wall of hollow carbon spheres is about 50 nm, and there are one or two nanoparticles in each carbon sphere. It can be seen from the Figure 1g that the nanoparticles and the inner wall of N/HCSs have close combination. In the high-resolution transmission electron microscopy (HRTEM) image, the lattice fringes of the nanoparticles with an interlayer distance of ≈0.253 nm correspond to the (311) plane of Fe_3O_4 (Figure 1g inset).[32] The corresponding EDS mapping images (Figure 1h–l) and line scan (Figure S6, Supporting Information) of Fe_{20}@N/HCSs show that Fe and O are mainly distributed on nanoparticles, while C and N are evenly distributed on the surface of N/HCSs. It can be indicated that the N/HCSs-encapsulated metal oxide nanoparticle materials with a yolk-shell structure are successfully prepared.

The phase structure of these obtained materials are further investigated by wide-angle X-ray diffraction (XRD), which shows that the metal oxide nanoparticles obtained belongs to Fe_3O_4 (JCPDS card No. 75-0033). (**Figure 2**a). With the increase of the ratio of Fe source, the XRD peaks of Fe_3O_4 become sharper, which is consistent with the TEM results. The graphitization degree of these samples are explored by Raman spectra (Figure 2b). The characteristic D (1350 cm^{-1}) and G (1590 cm^{-1}) bands are related to the disordered or defect carbon and graphitic sp^2-carbon, respectively.[33] With the increase of metal content, the I_D/I_G intensity ratio decreases gradually. This indicated the graphitization of carbon skeleton increases gradually, which is conducive to enhance the conductivity of these materials during electrocatalytic ORR/OER process.[34] The elemental chemical states of these samples are further analyzed by X-ray photoelectron spectroscopy (XPS) measurements. As shown in the Figure 2c, the high-resolution Fe 2p XPS for Fe_{20}@N/HCSs catalyst exhibits two types of Fe species: Fe(III) (711.5 and 724.4 eV) and Fe(II) (714.1 and 725.8 eV).[35] The characteristic signal at 718.7 eV is assigned to the satellite peak of Fe. The high-resolution XPS spectrum of Fe 2p is consistent with the XRD results, which indicated the presence of Fe_3O_4. The deconvolution of high-resolution XPS N 1s spectrum reveal the four types of nitrogen species in Fe_{20}@N/HCSs catalyst, including the pyridinic-N (398.6 eV), pyrrolic-N (399.7 eV), graphitic-N (400.9 eV) and oxidised-N (402.1 eV) (Figure 2d).[36,37] The nitrogen atom with higher electronegativity and electronic polarity breaks the original electronic structure balance of sp^2 hybrid in graphite layer, makes the heteroatom and the adjacent carbon atom charged, increases the density of active center, and thus improves the electrocatalytic ORR/OER activity.[11,38] Moreover, the formation of Fe–N_x can further increase the active sites, and decrease the energy barrier

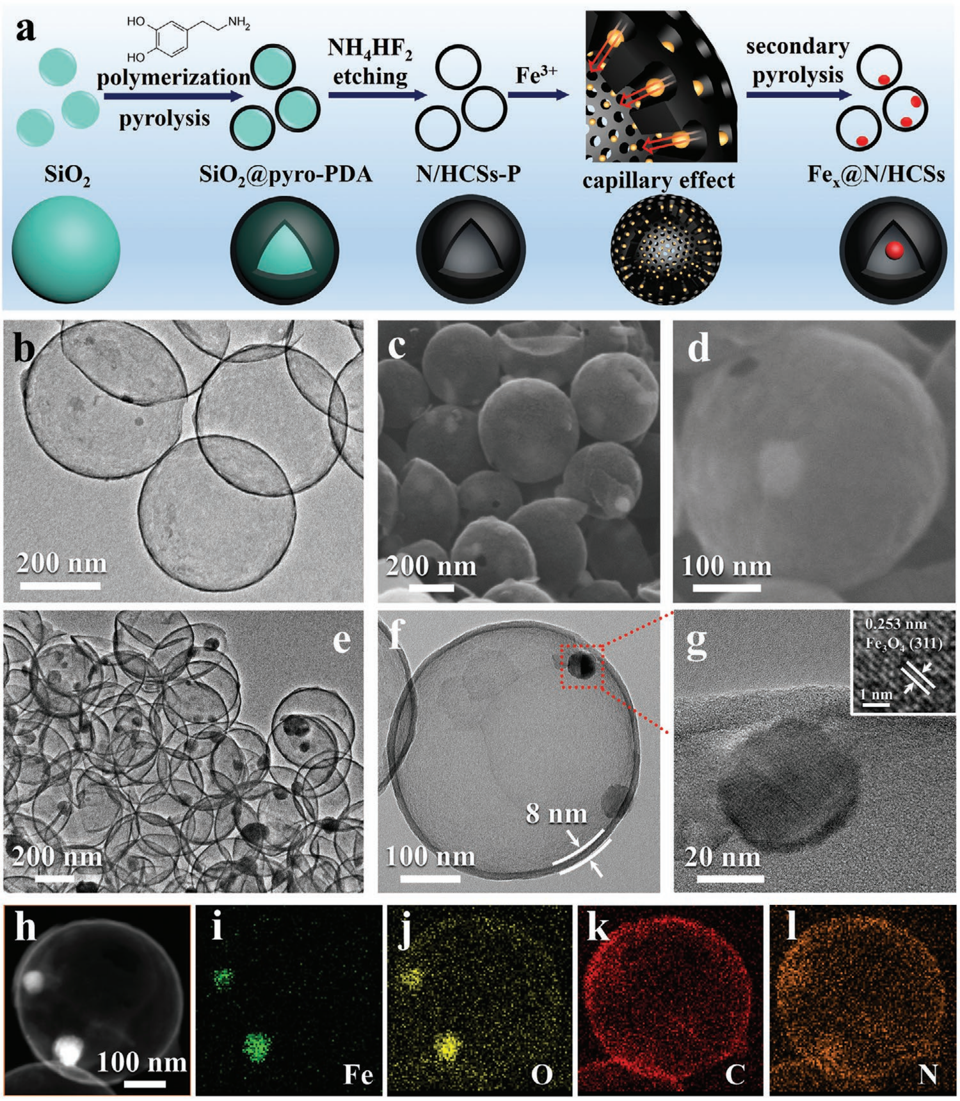

Figure 1. a) Schematic illustrating the synthesis procedure for Fe_x@N/HCSs catalysts. b) TEM image of N/HCSs-P. c,d) SEM, e,f) TEM, g) HR-TEM, and h–l) EDS mapping images of Fe_{20}@N/HCSs material.

of electrocatalytic ORR/OER.[39,40] In the high-resolution C 1s spectrum (Figure 2e), four characteristic peaks mainly assigned to the C=C, C=N&C−O, C−N&C−O and O−C=O.[35,41] And in the deconvolution of high-resolution O 1s spectrum shows the four types of oxygen species are attributed to the Fe−O, C=O&O−C=O and physically absorbed O (Figure 2f).[42,43] The surface and pore-size characterization of these Fe_x@N/HCSs materials further explored using N_2 adsorption–desorption test. As shown in the Figure S7a, Supporting Information, the BET specific surface area of these material decreases with the increase of Fe_3O_4 content, which is consistent with the TEM characterization. The increased Fe_3O_4 nanoparticles occupy more hollow carbon spherical cavities, resulting in the specific surface area decrease gradually. Combined with the subsequent electrochemical tests, the specific surface area is not the main factor affecting the electrocatalytic ORR/OER activity. The pore size distribution in all these investigated samples is calculated based on the Barrett–Joyner–Halenda model (Figure S7b, Supporting Information). The average pore diameter of these materials are about 4 nm that is assigned to the pore structure on the surface of N/HCSs, which signify the introduction of Fe_3O_4 nanoparticles will not destroy the pore structure of carbon shell.

The ORR performance of these prepared materials was investigated using rotating disk electrode techniques in 0.10 M KOH electrolyte. In the Figure S8, Supporting Information, the cyclic voltammetry curves of Fe_{20}@N/HCSs and commercial Pt/C reveal that both of them show the obvious reduction peak in the O_2-saturated electrolyte, while disappeared the electrochemical response peak in the N_2-saturated electrolyte. It indicated that these ORR activity is ascribed to the reduction of O_2.[33] As shown in the **Figure 3**a, the original N/HCSs show the worst ORR activity with the half-wave potential ($E_{1/2}$ = 0.740 V), and limited current density (J_L = 4.50 mA cm^{-2}). After the introduction of Fe_3O_4 nanoparticles, the ORR activity of these Fe_x@N/HCSs increased significantly, and Fe_{20}@N/HCSs

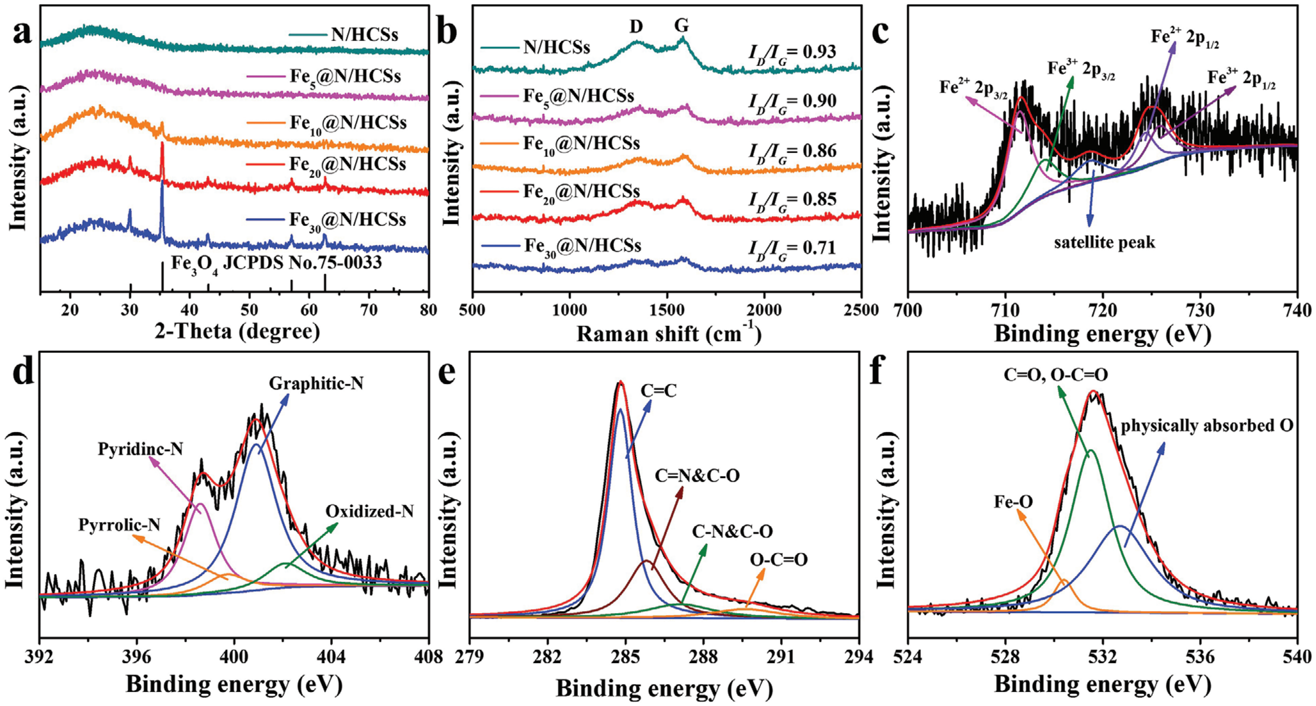

Figure 2. a) XRD pattern and b) Raman spectrum of N/HCSs and Fe_x@N/HCSs samples. High-resolution XPS spectra of c) Fe 2*p*, d) N 1s, e) C 1s, and f) O 1s for Fe_{20}@N/HCSs.

material exhibit the best performance ($E_{1/2}$ = 0.850 V and J_L = 5.750 mA cm^{-2}), which is comparable to that of the commercial 20 wt% Pt/C ($E_{1/2}$ = 0.859 V and J_L = 5.361 mA cm^{-2}), and also is outperforms these of many previously reported yolk-shell structure ORR electrocatalysts (Table S1, Supporting Information), such as Co NP/NC-700 ($E_{1/2}$ = 0.80 V and J_L = 5.10 mA cm^{-2}),[44] Co-C@Co_9S_8 DSNCs ($E_{1/2}$ = 0.83 V and J_L = 5.25 mA cm^{-2}),[45] Co/HNCP ($E_{1/2}$ = 0.845 V, Tafel plots = 64 mV dec^{-1}, and J_L = 5.401 mA cm^{-2}),[46] Fe@FeN_x/N-C ($E_{1/2}$ = 0.82 V, Tafel plots = 94.5 mV dec^{-1}, and J_L = 5.25 mA cm^{-2}),[47] Fe_3C@N/C-1 ($E_{1/2}$ = 0.815 V, Tafel plots = 68.62 mV dec^{-1}, and J_L = 5.78 mA cm^{-2}),[48] etc. Compared with the Fe_{20}@N/HCSs, the limited current density of Fe_{30}@N/HCSs become lower obviously, which attributed to the introduction of excess Fe_3O_4 nanoparticles reduce the electrical conductivity of electrocatalyst (Figure 3e), and the increased ratio of two electron transfer for excess Fe_3O_4 in the ORR.[49–51] Meanwhile, the Fe_{20}@N/HCSs exhibited the lowest Tafel slope of 58.7 mV dec^{-1} compared to those of N/HCSs (72.3 mV dec^{-1}), Fe_5@N/HCSs (65.8 mV dec^{-1}), Fe_{10}@N/HCSs (64.7 mV dec^{-1}), Fe_{30}@N/HCSs (62.1 mV dec^{-1}), and Pt/C (97.2 mV dec^{-1}), suggesting its superior kinetics for ORR (Figure 3b). The rotating ring-disc electrode carried out further reveal the electron transfer number (n) and hydrogen peroxide percentage for Fe_{20}@N/HCSs. As shown in the Figure 3c, the Fe_{20}@N/HCSs exhibits the low H_2O_2 yields (lower than 14%) and the electron transfer number near four at the potential ranging from 0.2 to 0.8 V, which is similar to that of commercial Pt/C. This result confirms the ORR on Fe_{20}@N/HCSs catalyst follows the effective 4e$^-$ pathway in alkaline medium. Moreover, the Fe_{20}@N/HCSs material reveals the excellent methanol resistance (Figure S9, Supporting Information). The stability of Fe_{20}@N/HCSs and Pt/C catalysts further evaluated via chronoamperometric measurements in O_2-saturated 0.1 м KOH solution (Figure 3d). The Fe_{20}@N/HCSs exhibits pronounced stability with 90.3% retention over 30 000 s of continuous operation, whereas Pt/C exhibits a faster current loss (72.7% retention). After the stability test, the structural morphology of Fe_{20}@N/HCSs was analyzed via TEM. As shown in the Figure S10, Supporting Information, Fe_{20}@N/HCSs still maintains the original yolk-shell structure, and Fe_3O_4 nanoparticles are still confined inside the N/HCSs, which indicates the material possess an excellent electrocatalytic ORR stability. Moreover, the electroconductivity of corresponding materials was assessed by electrochemical impedance spectroscopy measurements in 0.1 м KOH solution with the frequency range of 0.1–10 000 Hz. As shown in the Figure 3e, the N/HCSs and Fe_{20}@N/HCSs reveal the maximum and minimum diameter in the Nyquist plot, respectively. It indicates that Fe_{20}@N/HCSs has the optimum electron transfer rate,[52] which is consistent with the law of ORR activity for these synthesized electrocatalysts. It also proves that the introduction of excessive Fe_3O_4 will be detrimental to the conductivity of this yolk-shell carbon material, thus reducing the electrocatalytic ORR performance. According to the previously reported literature, the nitrogen doped in the carbon skeleton can coordinate with the transition metal to form the ORR active site.[53,54] In order to verify the existence of Fe–N_x active sites, 5 mmol KSCN was added into the electrolyte to complexate/poison this active sites.[55] As shown in the Figure 3f, after KSCN was added into the electrolyte, the ORR activity of material N/HCSs did not change, but the activity of material Fe_{20}@N/HCSs significantly deteriorated. All the initial potential, half wave potential, and

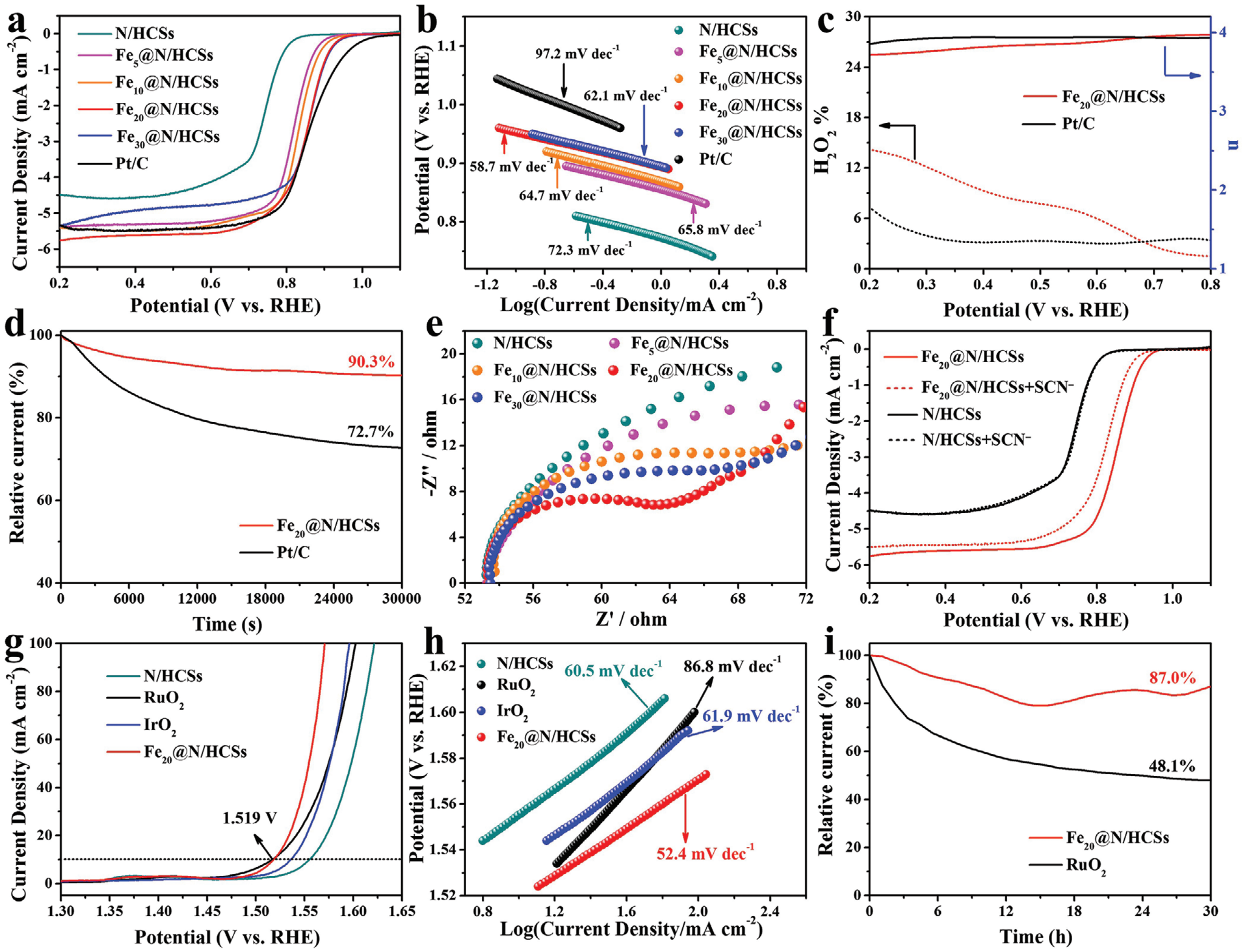

Figure 3. a) ORR LSV curves and b) corresponding Tafel plots of N/HCSs and Fe_x@N/HCSs. c) Percentage of peroxide and electron transfer number of Fe_{20}@N/HCSs and Pt/C. d) ORR i–t curves of Fe_{20}@N/HCSs and Pt/C. e) Nyquist plots of N/HCSs and Fe_x@N/HCSs catalysts-modified electrodes in 0.1 M KOH solution with the frequency range of 0.1–100 000 Hz, respectively. f) LSV curves of N/HCSs and Fe_{20}@N/HCSs with and without 5 mmol SCN^- in O_2-saturated 0.1 M KOH at a rotation rate of 1600 rpm with a scan rate of 5 mV s^{-1}. g) OER LSV curves and h) corresponding Tafel plots of N/HCSs, Fe_x@N/HCSs, RuO_2 and IrO_2. i) OER i–t curves of Fe_{20}@N/HCSs and RuO_2.

limiting current are attenuated, indicating the Fe–N_x coordination is important active site in this system.

The OER performance of these Fe_x@N/HCSs materials are further investigated in 1 M KOH electrolyte. Similar to ORR activity results, the Fe_{20}@N/HCSs also exhibits the best performance during the OER process (Figure S11, Supporting Information). Compared with the original N/HCSs (η_{10} = 325 mV), the overpotential of Fe_{20}@N/HCSs (η_{10} = 289 mV) at current density of 10 mA cm^{-2} is reduced by 36 mV, which is similar to commercial RuO_2 (η_{10} = 287 mV), exceeds commercial IrO_2 (η_{10} = 306 mV), and surpasses many previously reported yolk-shell structure OER electrocatalysts (Table S2, Supporting Information), such as yolk-shell Co-CoO/BC (η_{10} = 300 mV),[56] FeCo-C/N (η_{10} = 353 mV),[57] Porous yolk-shelled $MnCo_2O_4$ microrugbys (η_{10} = 360 mV),[58] Y-S Ni-Co-Se/CFP (η_{10} = 300 mV),[59] $Zn_xCo_{3-x}O_4$ YSP (η_{10} = 332 mV),[60] etc. Meanwhile, the corresponding Tafel plots of Fe_{20}@N/HCSs (52.4 mV dec^{-1}) also is minimum compared with those of RuO_2 (86.8 mV dec^{-1}), IrO_2 (61.9 mV dec^{-1}), and N/HCSs (60.5 mV dec^{-1}), indicating the Fe_{20}@N/HCSs also possesses superior kinetics for OER. In addition, the Fe_{20}@N/HCSs material retains a steady OER activity, and maintains high relative current of 87.0% after 30 h continuous chronoamperometry technique operation, whereas the commercial RuO_2 catalysts reveals a 51.9% decrease in activity under the same conditions. These electrochemical data reveal that Fe_{20}@N/HCSs can work as a desirable catalyst with superior activity and durability for ORR/OER, making it a promising air cathode electrocatalyst in a two-electrode primary Zn–air battery.

In order to evaluate the potential application of Fe_{20}@N/HCSs in energy storage and conversion devices, the Zn–air battery was assembled by applying Fe_{20}@N/HCSs loaded on the carbon fiber paper electrode as the air cathode, polished zinc foil as the anode, and 6 M KOH with 0.2 M zinc acetate electrolyte as the electrolyte (**Figure 4**a). For comparison, the Zn–air battery fabricated from commercial Pt/C and RuO_2 mixture

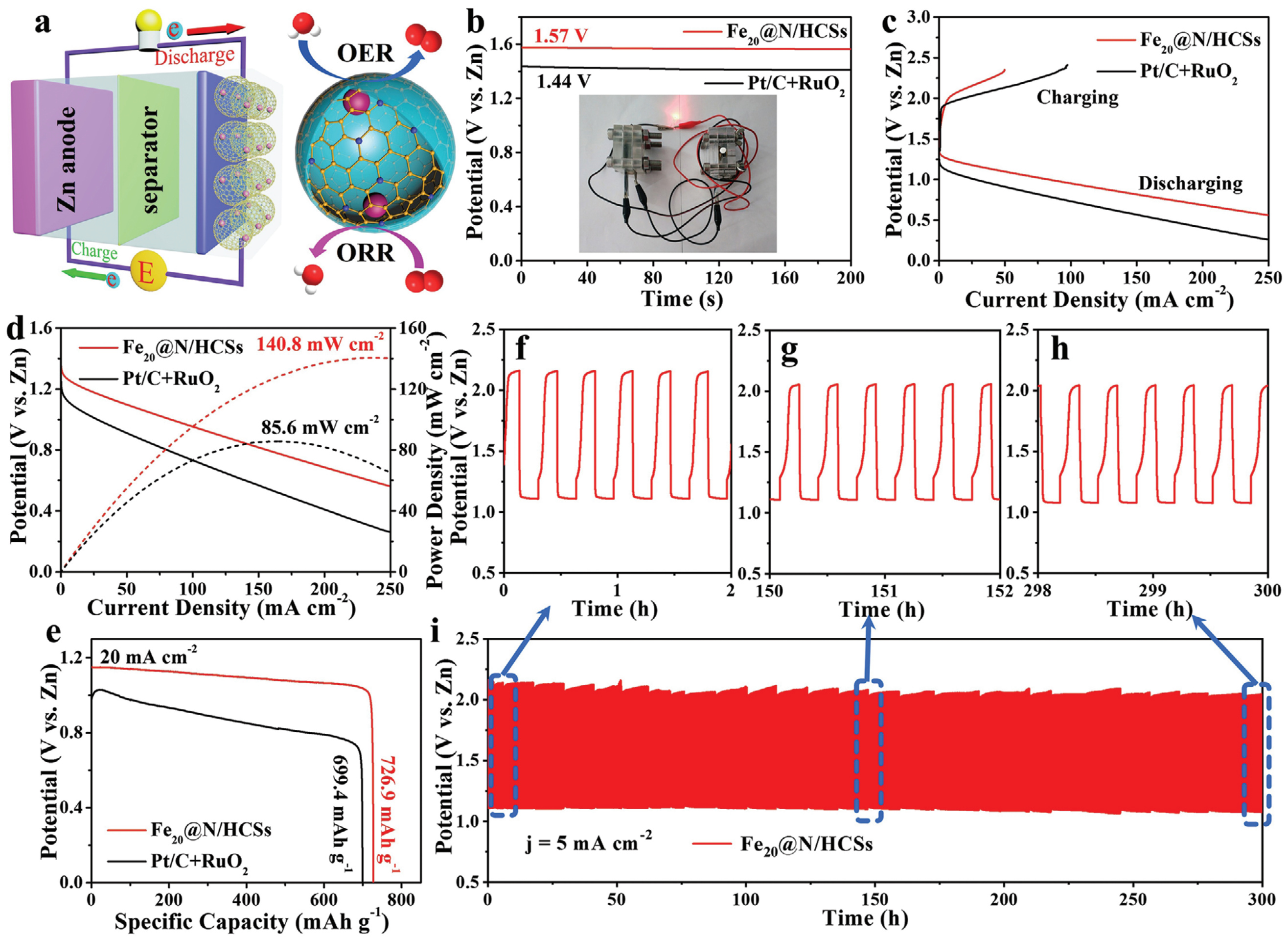

Figure 4. a) Schematic illustration of the aqueous Zn–air battery with Fe_{20}@N/HCSs as the air cathode. b) The open-circuit voltage (inset: the optical images of red LED lightened by Zn–air battery based on Fe_{20}@N/HCSs), c) charging–discharging, and d) power density curves e) discharge curves at a current density of 20 mA cm^{-2} of Zn–air battery based on Fe_{20}@N/HCSs and Pt/C+RuO_2 as air cathode. f–h) The amplified charge–discharge profiles upon cycling at different testing time. i) Cycling performance of the rechargeable Zn–air batteries based on Fe_{20}@N/HCSs as air cathode.

as the air cathode with the same amount of catalyst loading was also measured in an identical test condition. As shown in Figure 4b, the open circuit voltage of the Fe_{20}@N/HCSs (1.57 V) based battery is higher than that of Pt/C + RuO_2 (1.44 V) based battery, which suggests the Fe_{20}@N/HCSs based battery can output higher voltage. Two series-connected Zn–air batteries based on the Fe_{20}@N/HCSs catalyst can easily power up a red light-emitting diode (LED) (inset of Figure 4b). The typical rechargeable battery displays the charge and discharge polarization curves indicate the Fe_{20}@N/HCSs based battery possesses charge–discharge capabilities (Figure 4c). The maximum power density of Fe_{20}@N/HCSs based battery achieves as high as 140.8 mW cm^{-2} (Figure 4d), which outperforms that of Pt/C + RuO_2 based battery (85.6 mW cm^{-2}), and other previously reported carbon-based electrocatalysts (Table S3, Supporting Information). Meanwhile, the galvanostatic discharge curve of the battery with the Fe_{20}@N/HCSs air cathode exhibits higher specific capacity of 726.9 mA h g^{-1} than that of Pt/C + RuO_2 (699.4 mAh g^{-1}) at a discharge current density of 20 mA cm^{-2} (Figure 4e). The cyclic stability of Fe_{20}@N/HCSs cathode has been further investigated by galvanostatic discharge and charge test at the current density of 5 mA cm^{-2} (Figure 4f–i). The Fe_{20}@N/HCSs based battery exhibits better rechargeability with an initial charge potential of 2.09 V and a discharge potential of 1.10 V, indicating the charge–discharge voltage gap of Fe_{20}@N/HCSs based battery is lower than that of Pt/C + RuO_2 based battery (Figure S12, Supporting Information). Unlike the Pt/C + RuO_2 based battery with sharp voltage change after 30 h cycle test, the Fe_{20}@N/HCSs based battery exhibits outstanding long-term durability without obvious decay after charge–discharge cycle tests with 300 h (Figure 4f–i), which is far more than most of the reported zinc air battery materials. The above test demonstrates that Fe_{20}@N/HCSs based advanced rechargeable zinc–air battery with outstanding power density and long-term cyclic stability is promising to substitute noble-metal catalysts for zinc–air batteries.

Based on the abovementioned analysis and the recent research on yolk-shell structure materials for the ORR, OER, and rechargeable Zn–air battery, Fe_{20}@N/HCSs exhibits excellent electrocatalytic activity that is attributed to the following several key points. First, the special yolk-shell structure with confinement effect could promote the electrocatalytic oxygen reaction in single hollow sphere ecosystem and reduce the reaction delay caused by material agglomeration.[34,37] Second, the uniform small-size Fe_3O_4 nanoparticles are evenly embedded into the hollow carbon sphere cavity, which effectively improves the electrical conductivity of material, and strengthens the effective distribution of Fe–N_x active sites. The Fe–N_x active sites can simultaneously reduce the reaction energy barrier of electrocatalytic oxygen reaction, thus exhibiting excellent ORR/OER performance.[15,16,39] Third, constructing an oxygen reaction electrocatalyst with highly active and stable as the cathode of rechargeable zinc–air battery, the assembled battery could possess high open circuit voltage, large power density and significantly excellent long-term cycling performance.[3,4,61]

3. Conclusion

In summary, a facile synthetic method has been developed for the preparation of ultrathin mesoporous N/HCSs, which are then used as nanoreactors using the melting-diffusion strategy via capillary force to grow nanoscale Fe_3O_4 particles inside them to ultimately obtain Fe_{20}@N/HCSs yolk-shell structure composites. In the electrocatalysis ORR/OER process, the yolk-shell structure Fe_{20}@N/HCSs material can effectively inhibit the agglomeration of metal oxide nanoparticles, and the porous spherical reaction ecosphere is conducive to the electron transport, and accelerates the reaction rate. The construction of Fe–N_x active sites in the Fe_{20}@N/HCSs material cavity greatly improves the ORR ($E_{1/2}$ = 0.850 V and J_L = 5.750 mA cm^{-2})/OER (η_{10} = 289 mV) activity and stability, which can be compared with commercial grade Pt/C ($E_{1/2}$ = 0.859 V and J_L = 5.361 mA cm^{-2} for ORR), RuO_2 (η_{10} = 287 mV for OER), IrO_2 (η_{10} = 306 mV for OER) and many catalysts reported so far. Moreover, the zinc–air battery assembled with Fe_{20}@N/HCSs catalyst exhibits high open circuit voltage (1.57 V), large power density (140.8 mW cm^{-2}), considerable reversibility, and significantly excellent long-term cycling performance (over 300 h). This work provides new strategy for the exploration of inexpensive ORR/OER electrocatalysts with nanoscale confinement structure, which leads to significantly reduced manufacturing costs for zinc–air battery and improve their efficiency.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

B.W. and Y.Y. contributed equally to this work. This work was financially supported by the National Natural Science Foundation of China (No. 21576123, 21878134, and 21676128).

Conflict of Interest

The authors declare no conflict of interest.

Keywords

carbon spheres, Fe_3O_4 nanoparticles, space-confined, yolk-shell constructions, zinc–air batteries

Received: July 11, 2020
Revised: August 12, 2020
Published online:

[1] J. J. Xue, T. Wu, Y. Q. Dai, Y. N. Xia, *Chem. Rev.* **2019**, *119*, 5298.
[2] X. C. Yan, Y. Jia, X. D. Yao, *Chem. Soc. Rev.* **2018**, *47*, 7628.
[3] J. Fu, R. L. Liang, G. H. Liu, A. P. Yu, Z. Y. Bai, L. Yang, Z. W. Chen, *Adv. Mater.* **2019**, *31*, 1805230.
[4] J. Yi, P. C. Liang, X. Y. Liu, K. Wu, Y. Y. Liu, Y. G. Wang, Y. Y. Xia, J. J. Zhang, *Energy Environ. Sci.* **2018**, *11*, 3075.
[5] X. X. Wang, M. T. Swihart, G. Wu, *Nat. Catal.* **2019**, *2*, 578.
[6] J. Park, T. Kwon, J. Kim, H. Jin, H. Y. Kim, B. Kim, S. H. Joo, K. Lee, *Chem. Soc. Rev.* **2018**, *47*, 8173.
[7] Z. Z. Yuan, Y. B. Yin, C. X. Xie, H. M. Zhang, Y. Yao, X. F. Li, *Adv. Mater.* **2019**, 1902025.
[8] M. J. Wu, G. X. Zhang, M. H. Wu, J. Prakash, S. H. Sun, *Energy Storage Mater.* **2019**, *21*, 253.
[9] R. Cai, H. Y. Jin, D. Yang, K. T. Lin, K. Chan, J. K. Sun, Z. Chen, X. B. Zhang, W. H. Tan, *Nano Energy* **2020**, *71*, 104542.
[10] X. Q. Wang, Z. J. Li, Y. T. Qu, T. W. Yuan, W. Y. Wang, Y. E. Wu, Y. D. Li, *Chem* **2019**, *5*, 1486.
[11] L. Xu, Y. H. Tian, D. J. Deng, H. P. Li, D. Zhang, J. C. Qian, S. A. Wang, J. M. Zhang, H. N. Li, S. H. Sun, *ACS Appl. Mater. Interface* **2020**, *12*, 31340.
[12] X. F. Lu, B. Y. Xia, S. Q. Zang, X. W. Lou, *Angew. Chem., Int. Ed.* **2019**, *58*, 2.
[13] K. Zeng, X. J. Zheng, C. Li, J. Yan, J. H. Tian, C. Jin, P. Strasser, R. Z. Yang, *Adv. Funct. Mater.* **2020**, *27*, 2000503.
[14] S. Sultan, J. N. Tiwari, A. N. Singh, S. Zhumagali, M. R. Ha, C. W. Myung, P. Thangavel, K. S. Kim, *Adv. Energy Mater.* **2019**, *9*, 1900624.
[15] H. J. Shen, E. Gracia-Espino, J. Y. Ma, K. T. Zang, J. Luo, L. Wang, S. S. Gao, X. Mamat, G. Z. Hu, T. Wagberg, S. J. Guo, *Angew. Chem., Int. Ed.* **2017**, *56*, 13800.
[16] H. L. Fei, J. C. Dong, Y. X. Feng, C. S. Allen, C. Z. Wan, B. Volosskiy, M. F. Li, Z. P. Zhao, Y. L. Wang, H. T. Sun, P. F. An, W. X. Chen, Z. Y. Guo, C. Lee, D. L. Chen, I. Shakir, M. J. Liu, T. D. Hu, Y. D. Li, A. I. Kirkland, X. F. Duan, Y. Huang, *Nat. Catal.* **2018**, *1*, 63.
[17] H. M. Xu, S. Q. Ci, Y. C. Ding, G. X. Wang, Z. H. Wen, *J. Mater. Chem. A* **2019**, *7*, 8006.
[18] C. Wei, R. R. Rao, J. Y. Peng, B. T. Huang, I. E. L. Stephens, M. Risch, Z. C. Xu, S. H. Yang, *Adv. Mater.* **2019**, *31*, 1806296.
[19] X. J. Wei, Y. B. Zhang, B. K. Zhang, Z. Lin, X. P. Wang, P. Hu, S. L. Li, X. Tan, X. Y. Cai, W. Yang, L. Q. Mai, *Nano Energy* **2019**, *64*, 103899.
[20] H. Tian, J. Liang, J. Liu, *Adv. Mater.* **2019**, *31*, 190886.
[21] M. Z. Hou, Y. T. Qiu, G. Yan, J. M. Wang, D. Zhan, X. Liu, J. C. Gao, L. F. Lai, *Nano Energy* **2019**, *62*, 299.
[22] H. R. Chen, K. Shen, Y. P. Tan, Y. W. Li, *ACS Nano* **2019**, *13*, 7800.
[23] X. Wang, Y. Chen, Y. J. Fang, J. T. Zhang, S. Y. Gao, X. W. Lou, *Angew. Chem., Int. Ed* **2019**, *58*, 675.
[24] S. H. Wang, J. Teng, Y. Y. Xie, Z. W. Wei, Y. A. Fan, J. J. Jiang, H. P. Wang, H. G. Liu, D. W. Wang, C. Y. Su, *J. Mater. Chem. A* **2019**, *7*, 4036.
[25] C. Galeano, C. Baldizzone, H. Bongard, B. Spliethoff, C. Weidenthaler, J. C. Meier, K. J. J. Mayrhofer, *Adv. Funct. Mater.* **2014**, *24*, 220.
[26] L. S. Lin, X. Y. Yang, Z. J. Zhou, Z. Yang, O. Jacobson, Y. J. Liu, A. Yang, G. Niu, J. B. Song, H. H. Yang, X. Y. Chen, *Adv. Mater.* **2017**, *29*, 1606681.

[27] Z. Li, Y. J. Fang, J. T. Zhang, X. W. Lou, *Adv. Mater.* **2018**, *30*, 1800525.
[28] W. W. Sun, C. Liu, Y. J. Li, S. Q. Luo, S. K. Liu, X. B. Hong, K. Xie, Y. M. Liu, X. J. Tan, C. M. Zheng, *ACS Nano* **2019**, *13*, 12137.
[29] X. E. Zhang, R. F. Zhao, Q. H. Wu, W. L. Li, C. Shen, L. B. Ni, H. Yan, G. W. Diao, M. Chen, *ACS Nano* **2017**, *11*, 8429.
[30] Q. Li, J. N. Guo, H. Zhu, F. Yan, *Small* **2019**, *15*, 1804874.
[31] N. Li, Q. Zhang, J. Liu, J. Joo, A. Lee, Y. Gan, Y. Yin, *Chem. Commun.* **2013**, *49*, 5135.
[32] G. P. Liu, B. Wang, P. H. Ding, Y. Z. Ye, W. X. Wei, W. S. Zhu, L. Xu, J. X. Xia, H. M. Li, *J. Alloys Compd.* **2019**, *797*, 849.
[33] B. Wang, L. Xu, G. P. Liu, P. F. Zhang, W. S. Zhu, J. X. Xia, H. M. Li, *J. Mater. Chem. A* **2017**, *5*, 20170.
[34] J. Y. Liu, H. Xu, H. P. Li, Y. H. Song, J. J. Wu, Y. J. Gong, L. Xu, S. Q. Yuan, H. M. Li, P. M. Ajayan, *Appl. Catal., B* **2019**, *243*, 151.
[35] L. S. Xie, X. L. Li, B. Wang, J. Meng, H. T. Lei, W. Zhang, R. Cao, *Angew. Chem., Int. Ed.* **2019**, *58*, 18883.
[36] G. P. Liu, B. Wang, P. H. Ding, W. X. Wei, Y. Z. Ye, L. Wang, W. S. Zhu, H. M. Li, J. X. Xia, *J. Alloys Compd.* **2020**, *815*, 152470.
[37] J. Y. Liu, L. Xu, Y. L. Deng, X. W. Zhu, J. J. Deng, J. B. Lian, J. J. Wu, J. C. Qian, H. Xu, S. Q. Yuan, H. M. Li, P. M. Ajayan, *J. Mater. Chem. A* **2019**, *7*, 14291.
[38] C. Tang, Q. Zhang, *Adv. Mater.* **2017**, *29*, 1604103.
[39] Y. J. Chen, S. F. Ji, Y. G. Wang, J. C. Dong, W. X. Chen, Z. Li, R. A. Shen, L. R. Zheng, Z. B. Zhuang, D. S. Wang, Y. D. Li, *Angew. Chem., Int. Ed* **2017**, *56*, 6937.
[40] P. Z. Chen, T. P. Zhou, L. L. Xing, K. Xu, Y. Tong, H. Xie, L. D. Zhang, W. S. Yan, W. S. Chu, C. Z. Wu, Y. Xie, *Angew. Chem., Int. Ed* **2017**, *56*, 610.
[41] B. Wang, J. Di, L. Lu, S. C. Yan, G. P. Liu, Y. Z. Ye, H. T. Li, W. S. Zhu, H. M. Li, J. X. Xia, *Appl. Catal., B* **2019**, *254*, 551.
[42] M. T. Qiao, X. F. Lei, Y. Ma, L. D. Tian, X. W. He, K. H. Su, Q. Y. Zhang, *Nano Res.* **2018**, *11*, 1500.
[43] J. Zhang, Y. J. Ji, P. T. Wang, Q. Shao, Y. Y. Li, X. Q. Huang, *Adv. Funct. Mater.* **2019**, *30*, 1906579.
[44] J. Zhao, F. Rong, Y. Ya, W. J. Fan, M. R. Li, Q. H. Yang, *J. Energy Chem.* **2018**, *27*, 1261.
[45] H. Hu, L. Han, M. Z. Yu, Z. Y. Wang, X. W. Lou, *Energy Environ. Sci.* **2016**, *9*, 107.
[46] D. N. Ding, K. Shen, X. D. Chen, H. R. Chen, J. Y. Chen, T. Fan, R. F. Wu, Y. W. Li, *ACS Catal.* **2018**, *8*, 7879.
[47] Y. J. Chen, Z. H. Shi, Z. H. Wang, C. Wang, J. G. Feng, B. L. Pang, Q. Sun, L. Y. Yu, L. F. Dong, *J. Alloys Compd.* **2020**, *829*, 154558.
[48] M. Chen, Y. Jiang, P. Mei, Y. Zhang, X. F. Zheng, W. Xiao, Q. L. You, X. M. Yan, H. L. Tang, *Polymers* **2019**, *11*, 767.
[49] X. F. Zhang, X. Y. Wang, L. J. Le, A. Ma, S. Lin, *J. Mater. Chem. A* **2015**, *3*, 19273.
[50] H. T. Wang, W. Wang, Y. Y. Xu, S. Dong, J. W. Xiao, F. Wang, H. F. Liu, B. Y. Xia, *ACS Appl. Mater. Interfaces* **2017**, *9*, 10610.
[51] Y. Wang, M. M. Wu, K. Wang, J. W. Chen, T. W. Yu, S. Q. Song, *Adv. Sci.* **2020**, *7*, 2000407.
[52] Z. Mo, X. W. Zhu, Z. F. Jiang, Y. H. Song, D. B. Liu, H. P. Li, X. F. Yang, Y. B. She, Y. C. Lei, S. Q. Yuan, H. M. Li, L. Song, Q. Y. Yan, H. Xu, *Appl. Catal. B-Environ* **2019**, *256*, 117854.
[53] A. A. Gewirth, J. A. Varnell, A. M. DiAscro, *Chem. Rev* **2018**, *118*, 2313.
[54] D. F. Yan, Y. X. Li, J. Huo, R. Chen, L. M. Dai, S. Y. Wang, *Adv. Mater.* **2017**, *29*, 1606459.
[55] H. B. Tan, J. Tang, J. Kim, Y. V. Kaneti, Y. M. Kang, Y. Sugahara, Y. Yamauchi, *J. Mater. Chem. A* **2019**, *7*, 1380.
[56] M. J. Yang, D. F. Wu, D. J. Cheng, *Int. J. Hydrogen Energy* **2019**, *44*, 6525.
[57] C. C. Zhang, H. Yang, D. Zhong, Y. Xu, Y. Z. Wang, Q. Yuan, Z. Z. Liang, B. Wang, W. Zhang, H. Q. Zheng, T. Cheng, R. Cao, *J. Mater. Chem. A* **2020**, *8*, 9536.
[58] J. Li, Y. X. Zhang, L. Li, Y. M. Wang, L. Zhang, B. J. Zhang, F. Wang, B. Li, X. Y. Yu, *Dalton Trans.* **2019**, *48*, 17022.
[59] K. L. Ao, J. C. Dong, C. H. Fan, D. Wang, Y. B. Cai, D. W. Li, F. L. Huang, Q. F. Wei, *ACS Sustainable Chem. Eng.* **2018**, *6*, 10952.
[60] Z. Yu, Y. Bai, Y. X. Liu, S. M. Zhang, D. D. Chen, N. Q. Zhang, K. N. Sun, *ACS Appl. Mater. Interfaces* **2017**, *9*, 31777.
[61] H. Q. Ji, M. F. Wang, S. S. Liu, H. Sun, J. Liu, T. Qian, C. L. Yan, *Energy Storage. Mater.* **2020**, *27*, 226.